

ST. FRANCIS INSTITUTE OF TECHNOLOGY
DEPARTMENT OF INFORMATION TECHNOLOGY
SECURITY LAB

Experiment – 7: Study the use of Passive Network Reconnaissance tools

Aim: To study the use of passive network reconnaissance tools, such as WHOIS, dig, traceroute, nslookup, etc. to gather information about networks and domain registrars.

Objective: After performing the experiment, the students will be able to apply basic network commands to gather network information.

Lab objective mapped: L502.6: Students should be able to apply network security basics, analyse different attacks on networks and evaluate the performance of firewalls and security protocols, such as SSL, IPSEC, and PGP, and authentication mechanisms to design secure applications.

Prerequisite: Basic knowledge of passive attack.

Requirements: Ubuntu/Unix/Linux Operating system

Pre-Experiment Theory:

A. Passive Reconnaissance through network commands

1. **WHOIS:** WHOIS is the Linux utility for searching an object in a WHOIS database. WHOIS is a database of domains, which includes a publicly displayed information about domains ownership, billing, technical, administrative, and nameserver information.

Running a WHOIS on your domain will look the domain up at the registrar for the domain information. All domains have WHOIS information. WHOIS database can be queried to obtain the following information,

- Administrative contact details, including names, email addresses, and telephone numbers.
- Mailing addresses for office locations relating to the target organization.
- Details of authoritative name servers for each given domain.

Example: `$ whois example.com` *(Use any URL of your choice)*

2. **Dig (Domain Information Groper):** Dig is a networking tool that can query DNS servers for information. It is very helpful for diagnosing problems with domain pointing and is a good way to verify that your configuration is working. The most basic way to use dig is to specify the domain you wish to query.

Example: `$ dig www.example.com` *(Use any URL of your choice)*

3. **Traceroute** - traceroute prints the route that packets take to a network host. Traceroute utility uses the TTL field in the IP header to achieve its operation. For users who are new to the TTL field, this field describes how much hops a particular packet will take while traveling on the network. So, this effectively outlines the lifetime of the packet on the network. This field is usually set to 32 or 64. Each time the packet is held on an intermediate router, it decreases the TTL value by 1. When a router finds the TTL value of 1 in a received packet then that packet is not forwarded but instead discarded. After discarding the packet, the router sends an ICMP error message of —Time exceeded back to the source from where the packet was generated. The ICMP packet that is sent back contains the IP address of the router. So now it can be easily understood that traceroute operates by sending packets with TTL value starting from 1 and then incrementing by one each time. Each time a router receives the packet, it checks the TTL field, if TTL field is 1 then it discards the packet and sends the ICMP error packet containing its IP address and this is what traceroute requires. So traceroute incrementally fetches the IP of all the routers between the source and the destination.

Example: `$ traceroute example.com` *(Use any URL of your choice)*

4. **Nslookup** - The nslookup command is used to query internet name servers interactively for information. nslookup, which stands for "name server lookup", is a useful tool for finding out information about a named domain. By default, nslookup will translate a domain name to an IP address (or vice versa).

Example: \$ nslookup example.com *(Use any URL of your choice)*

B. Passive Reconnaissance through publicly available tools

1. **archive.org** (<https://archive.org/>)

In the archive.org website we can get the complete history of any website like when it was last updated. We can go back to a particular date and observe the webpage. We can mirror the website which will load all the files locally, such as HTML codes, images etc. that can be used to observe the directories used.

2. **Whois** (<https://www.whois.com/>)

Whois database lookup allows us to access many useful information about target such as:

- Registration details
- IP address
- Contact number and Email ID
- Domain owner
- Name servers
- Regional Internet Registries

3. **Netcraft site report** (<https://sitereport.netcraft.com/>)

Netcraft is an internet service organization, used to collect information such as IP address, services running on systems, operating systems, name servers, technologies used by websites.

Procedure & Outputs:

1. With Linux/Ubuntu/Unix operating systems run the commands discussed in part A of theory section. Analyze the output. Take screenshots (SS). Describe your observations under each SS in detail. Use indicators such as highlight, colour, and box for this purpose.
2. Browse the web tools discussed in part B of the theory section. Identify following:
 - a. Using 'archive.org' find the update history of 'sfit.ac.in' domain.
 - b. Perform a passive reconnaissance using the Calendar, Changes, Summary, Site Map, URL tabs. Take appropriate screenshots. Describe your observations under each SS in detail. Use indicators such as highlight, colour, and box for this purpose.
 - c. Using 'whois.com' find the domain information of 'facebook.com'. Take appropriate screenshots. Indicate the following information in your screenshots and complete the observation table given in the observation section.
 - d. Using 'netcraft.com' find the site report of 'microsoft.com'. Perform passive reconnaissance for useful information. Take appropriate screenshots. Describe your observations under each SS in detail. Use indicators such as highlight, colour, and box for this purpose. Complete the observation table given in the observation section.

Post Experimental Exercise Questions:

1. What is network reconnaissance?
2. What is passive reconnaissance? Give some examples.
3. What is active reconnaissance? Give some examples.

Conclusion:

In this experiment we studied various reconnaissance tools that can be used to gather primary information about the target/victim before launching any cyber-attack.

References:

1. "How to Use Linux dig Command", <https://phoenixnap.com/kb/linux-dig-command-examples>
2. "Lecture 17: Information Gathering (Part 1)", <https://youtu.be/mLvwpiR4dG4>

The screenshot shows the whois.com website interface. At the top, there's a navigation bar with links for Domains, Hosting, Websites, Email, Security, Whois, and Deals. The main heading is 'chatgpt.com' with a 'Make offer' button. Below this, a 'Domain Information' box contains the following details:

Domain:	chatgpt.com
Registered On:	2022-11-30
Expires On:	2026-11-30
Updated On:	2024-10-17
Status:	client delete prohibited client transfer prohibited client update prohibited server delete prohibited server transfer prohibited server update prohibited
Name Servers:	hassan.ns.cloudflare.com savanna.ns.cloudflare.com

Observation: The WHOIS lookup shows that chatgpt.com is registered with MarkMonitor and has an expiry date of 30 November 2026. The domain is protected by multiple client and server status restrictions, indicating that transfers and updates are restricted.

```
student@Demita:~/Shirley$ whois chatgpt.com
Domain Name: CHATGPT.COM
Registry Domain ID: 2741883935_DOMAIN_COM-VRSN
Registrar WHOIS Server: whois.markmonitor.com
Registrar URL: http://www.markmonitor.com
Updated Date: 2024-10-17T22:20:15Z
Creation Date: 2022-11-30T23:59:18
Registry Expiry Date: 2026-11-30T23:59:19Z
Registrar: MarkMonitor Inc.
Registrar IANA ID: 292
Registrar Abuse Contact Email: abusecomplaints@markmonitor.com
Registrar Abuse Contact Phone: +1.2086851750
Domain Status: clientDeleteProhibited https://icann.org/epp#clientDeleteProhibited
Domain Status: clientTransferProhibited https://icann.org/epp#clientTransferProhibited
Domain Status: clientUpdateProhibited https://icann.org/epp#clientUpdateProhibited
Domain Status: serverDeleteProhibited https://icann.org/epp#serverDeleteProhibited
Domain Status: serverTransferProhibited https://icann.org/epp#serverTransferProhibited
Domain Status: serverUpdateProhibited https://icann.org/epp#serverUpdateProhibited
Name Server: HASSAN.NS.CLOUDFLARE.COM
Name Server: SAVANNA.NS.CLOUDFLARE.COM
DNSSEC: unsigned
URL of the ICANN Whois Inaccuracy Complaint Form: https://www.icann.org/wicf/
>>> Last update of whois database: 2026-09-21T06:16:49Z <<<
```

Observation: The whois chatgpt.com command provides detailed domain information such as the registrar, creation date, expiry date, domain status, and name servers. It also shows that the domain uses Cloudflare name servers and that DNSSEC is unsigned.

```
student@Demita:~/Shirley$ dig www.chatgpt.com

; <<>> DiG 9.18.39-0ubuntu0.22.04.3-Ubuntu <<>> www.chatgpt.com
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 1751
;; flags: qr rd ra; QUERY: 1, ANSWER: 2, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
;; EDNS: version: 0, flags:; udp: 65494
;; QUESTION SECTION:
;www.chatgpt.com.                IN      A

;; ANSWER SECTION:
www.chatgpt.com.                300     IN      A      172.64.155.209
www.chatgpt.com.                300     IN      A      104.18.32.47

;; Query time: 7 msec
;; SERVER: 127.0.0.53#53(127.0.0.53) (UDP)
;; WHEN: Mon Sep 21 11:48:58 IST 2026
;; MSG SIZE rcvd: 76
```

Observation: The dig command resolves www.chatgpt.com to two IPv4 addresses: 172.64.155.209 and 104.18.32.47. The DNS query completed successfully with a response time of approximately 7 ms.

```
student@Demita:~/Shirley$ traceroute chatgpt.com
traceroute to chatgpt.com (104.18.32.47), 30 hops max, 60 byte packets
 1 sophos.sfit.co.in (192.168.7.254)  0.604 ms  0.564 ms  0.547 ms
 2 * static-125.131.252.123-tataidc.co.in (123.252.131.125)  1.907 ms *
 3 10.129.10.226 (10.129.10.226)  7.640 ms  4.124 ms *
 4 * * as13335.bom.extreme-ix.net (103.77.108.118)  42.558 ms
 5 162.158.226.89 (162.158.226.89)  5.138 ms 162.158.226.73 (162.158.226.73)
   4.629 ms 162.158.226.93 (162.158.226.93)  7.184 ms
 6 104.18.32.47 (104.18.32.47)  3.260 ms  5.708 ms  5.061 ms
```

Observation: The traceroute shows the network path from the system to chatgpt.com, passing through several intermediate routers before reaching 104.18.32.47. Some hops show *, indicating that those routers did not respond to traceroute probes, while the final destination responded successfully.

```
student@Demita:~/Shirley$ nslookup chatgpt.com
Server:          127.0.0.53
Address:         127.0.0.53#53

Non-authoritative answer:
Name:   chatgpt.com
Address: 104.18.32.47
Name:   chatgpt.com
Address: 172.64.155.209
Name:   chatgpt.com
Address: 2a06:98c1:3101::6812:202f
Name:   chatgpt.com
Address: 2606:4700:4408::ac40:9bd1
```

Observation: The nslookup command provides the DNS information for chatgpt.com, including its resolved IP addresses and DNS server details. It confirms that the domain is successfully resolving through the configured DNS server.

Target Domain/URL/Website for whois : chatgpt.com			
Registrar:	MarkMonitor Inc.	Registration Expiry date:	2026-11-30
Registration Update date:	2024-10-17	Name Servers:	hassan.ns.cloudflare.com savanna.ns.cloudflare.com
Registrant Organization	OpenAI	Registrant Country:	US

archive.org

1)

Calendar

Collections

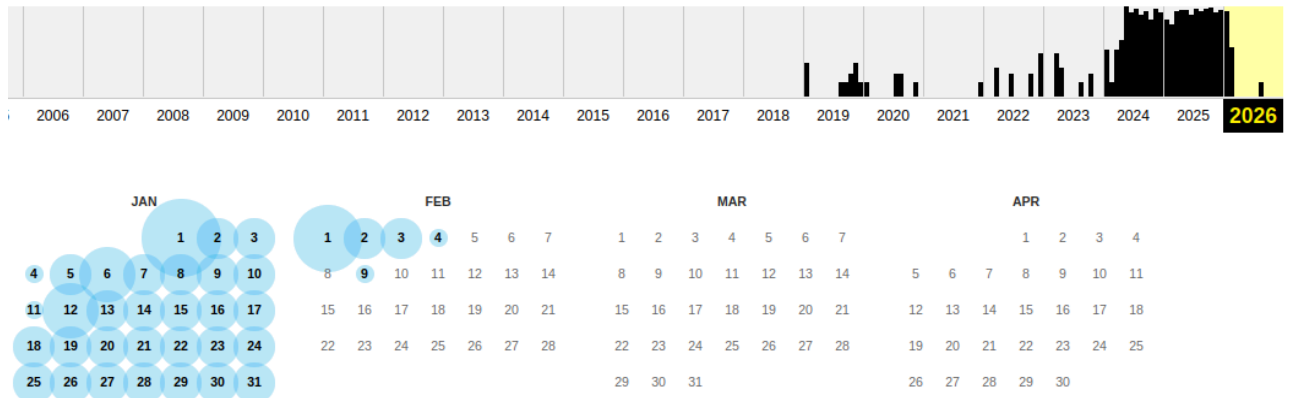
Changes

Summary

Site Map

URLs

Saved **1,406 times** between **January 2, 2019** and **August 31, 2026**.



Calendar: The Wayback Machine shows that the SFIT website was archived **1,406 times** from **2019** to **2026**, indicating regular captures over the years.

2)

Calendar

Collections

Changes

Summary

Site Map

URLs

Saved **1,406 times** between **January 2, 2019** and **August 31, 2026**.



No collection information available for 2026.

The Wayback Machine doesn't have any collection information available for 2026.

Do you want to list all archived URLs with this prefix: www.sfit.ac.in?

Collections: No collection information is available for **2026**, although archived URLs for the SFIT website can still be accessed.

3)

Calendar

Collections

Changes

Summary

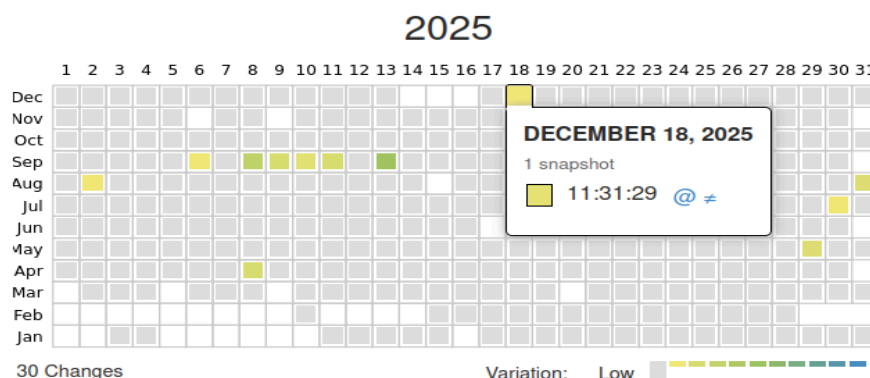
Site Map

URLs

"Changes" is a tool you can use to identify, and display, changes in the content of archives of URLs.

First you can select two different archives for a URL, based on an interface that shows the degree of relative change from one archive to another.

Then you can see the replay of the two URLs you select, side-by-side, with changes highlighted in Blue and Yellow.



Changes: The Changes section shows multiple snapshots in **2025**, allowing comparison of how the website content changed over time.

4)

host www.sfit.ac.in

Indexed on February 12, 2026.

See also: Domain Summary: www.sfit.ac.in · Parent Domain Summary: sfit.ac.in · TLD Summary: ac.in

Saved **1,406** times between January 2, 2019 and August 31, 2026.

MIME-types

Any MIME-types

Year Start

2019

Year End

2026

Quick Filter

All

image

text

application

font

video

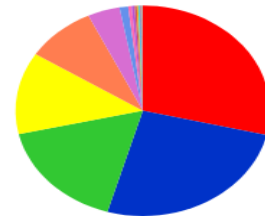
Summary on MIME-types Count

Quick search on MIME-types...

<< < 1 2 > >>

Captures

	Captures	Unique URLs
text/html	2.4K	329
image/jpeg	2.1K	802
image/png	1.4K	367
application/pdf	1K	718
text/css	738	39
application/javascript	338	34
image/gif	91	13
font/woff2	36	4
application/vnd.ms-excel	31	31
application/vnd.ms-fontobject	23	4



Summary: The summary shows different MIME types and their capture counts, with **HTML**, **JPEG**, **PNG**, and **PDF** among the most frequently archived content types.

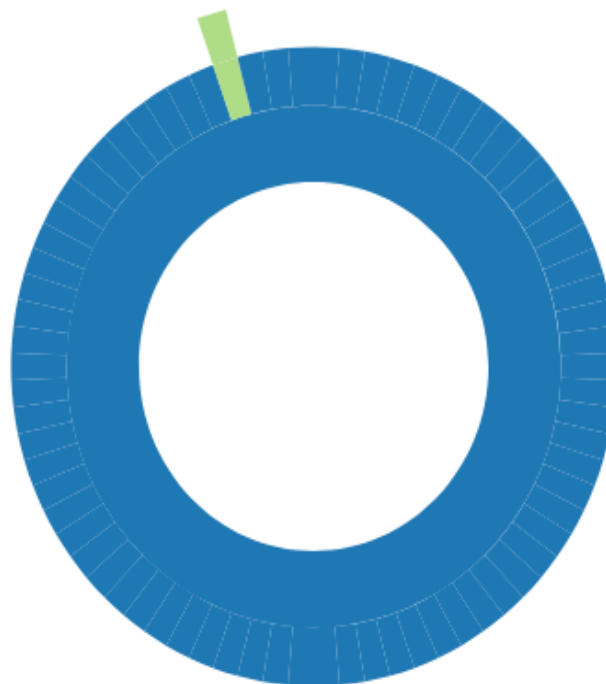
5)

host www.sfit.ac.in

This "Site Map" feature groups all the archives we have for websites by year, then builds a visual site map, in the form of a radial-tree graph, for each year.

The center circle is the "root" of the website and successive rings moving out from the center present pages from the site.

As you roll-over the rings and cells note the corresponding URLs change at the top, and that you can click on any of the individual pages to go directly to an archive of that URL

2019 2020 2021 2022 2023 2024 **2025** 2026

Top unique sub-pages

- /aac%20mom.php (1)
- /about-sfit.php (1)
- /academic%20calen... (1)
- /academic-researc... (1)
- /achievements.php (1)
- /administration-c... (1)
- /admissionapprova... (1)
- /aichte-approval.p... (1)
- /aiml.php (1)
- /alumni-testimoni... (1)
- /aqar.php (1)
- /bachelor-enginee... (1)
- /be_exam_2020.php (1)
- /career_sfit.php (1)
- /certification-co... (1)
- /change-nomenclat... (1)
- /cmpn.php (1)
- /college-emblem-a... (1)
- /conception%20202... (1)
- /conduct-ethic-st... (1)

Site Map: The Site Map displays the archived SFIT website structure across different years and shows the various webpages captured under the domain.

6)

2,404 URLs have been captured for this URL prefix.

Filter results by URL or MIME Type (i.e. ".txt")							
URL	MIME Type	From	To	Captures	Duplicates	Uniques	
http://sfit.ac.in:80/library.php	text/html	Mar 23, 2019	Aug 28, 2025	19	9	10	
https://sfit.ac.in/pdf/IQAC/IQAC Vision-Objectives-Functions-Strategies (1) (2).pdf	application/pdf	Dec 23, 2022	Aug 26, 2025	7	6	1	
https://sfit.ac.in/sfitengadmin/skin/css/bootstrap.min.css	text/css	May 4, 2024	May 4, 2024	1	0	1	
https://sfit.ac.in/sfitengadmin/skin/css/plugins/metisMenu/metisMenu.min.css	text/css	May 4, 2024	May 4, 2024	1	0	1	
https://sfit.ac.in/sfitengadmin/skin/css/sb-admin-2.css	text/css	May 4, 2024	May 4, 2024	1	0	1	

URLs: The URLs section shows that **2,404 URLs** have been captured for the SFIT website, along with their MIME types, capture dates, and number of duplicates.

netcraft.com

Research Tools

Internet Intelligence

Platform Overview

REPORT FRAUD ↗

Background

Site title	Not Present	Date first seen	February 2023
Site rank	528606	Primary language	English
Description	Not Present		

Network

Site	http://www.chatgpt.com ↗	Domain	chatgpt.com
Netblock Owner	Cloudflare, Inc.	Nameserver	hassan.ns.cloudflare.com
Hosting company	Cloudflare	Domain registrar	markmonitor.com
Hosting country	US	Nameserver organisation	whois.cloudflare.com
IPv4 address	104.18.32.47 (VirusTotal ↗)	Organisation	OpenAI, United States
IPv4 autonomous systems	AS13335 ↗	DNS admin	dns@cloudflare.com
IPv6 address	2606:4700:4408:0:0:ac40:9bd1	Top Level Domain	Commercial entities (.com)
IPv6 autonomous systems	AS13335 ↗	DNS Security Extensions	Enabled
Reverse DNS	Unknown		

Netcraft shows that chatgpt.com uses Cloudflare for hosting/network services, with the domain registered through MarkMonitor and DNS man

Target Domain/URL/Website for netcraft : chatgpt.com			
IPv4 address:	72.64.155.209	SSL/TLS certificate Issuing organization:	Google Trust Services
Certificate Validity period:	From Sep 16 2026 to Dec 15 2026 (2 months, 4 weeks, 1 day)	Public key algorithm:	id-ecPublicKey
Public key length:	256	Certificate Hash:	fQCfAfOY50vsEHEOEB40 xEFxgU0
Signature algorithm:	ecdsa-with-SHA256	Public Key Hash:	dfaba3e1ae23317a2e48eea2 492df60897865b9efe0ec8e23 7c90d4f079c3324
Server-Side site technology:	Not present	Client-Side site technology:	JavaScript Client Pull